

Multiple Choice (10 marks)

1. What type of diet is recommended for weight maintenance?
 - (a) Low protein and low GI
 - (b) High protein and high GI
 - (c) Low protein and high GI
 - (d) High protein and low GI
2. Which of the following vitamins provides the coenzyme for oxidative decarboxylation of pyruvate?
 - (a) Folate
 - (b) Niacin
 - (c) Riboflavin
 - (d) Thiamin
3. Which of the following statements is correct (according to knowledge in 2020)?
 - (a) Consumers with phenylketonuria must avoid the consumption of the sweetener aspartame
 - (b) Consumers with phenylketonuria must avoid the consumption of the sweetener saccharin
 - (c) Consumers with phenylketonuria must avoid the consumption of the sweetener sucralose
 - (d) Consumers with phenylketonuria must avoid the consumption of the sweetener acesulfame K
4. Oral bacterial synthesise extracellular glucans (dextran and mutan) from which host dietary component?
 - (a) glucose
 - (b) Alpha limit dextrans
 - (c) Sucrose
 - (d) polysaccharides
5. Which vitamins are important in lowering circulating homocysteine levels
 - (a) Vitamin D
 - (b) Vitamin C
 - (c) Vitamin A
 - (d) Folate, vitamins B₆ and B₁₂

True / False (10 marks)

Answer True or False

6. True or False: Rising incomes and urbanization globally are associated with an increased consumption of meat and dairy products.
7. True or False: The phenotype of a congenital disorder impairing leptin secretion includes obesity, abnormal growth, hypothyroidism, and hyperinsulinaemia.
8. True or False: Adaptive thermogenesis is the body's way of adjusting heat production in response to changes in nutrition.
9. True or False: Thiamin deficiency in alcoholics primarily causes Cushing syndrome, a disorder related to cortisol levels.
10. True or False: Plasma lipoproteins have a hydrophobic core composed of triacylglycerol and cholesterol esters.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the roles of intestinal "brakes" and "accelerators" in regulating digestion, and analyze why the statement about dietary starch and insulin release might be considered false.
12. Discuss the role of mitochondria and peroxisomes in fatty acid β -oxidation, and explain why the statement that peroxisomes are the major site of this process is incorrect.

End of Paper